INTEGRATION BEE 2025

Meia-final 1

1. $$\int_0^{2025} \lfloor \sqrt{x} \rfloor dx = 59730$$

2. $$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \max\left\{2\sin^2(x), 2\cos^2(x)\right\} dx = \pi + 2$$

3. $$\int_0^1 \frac{1}{\sqrt{|\log(x)|}} dx = \sqrt{\pi}$$

4. $$\int_{-\infty}^{+\infty} x^{100} e^{-x^2} dx = \frac{\sqrt{\pi}}{4^{50}} \frac{100!}{50!}$$

Meia-final 2

1. $$\int_0^{\infty} \frac{1}{(x^2+4)^2} dx = \frac{\pi}{32}$$

2. $$\int_0^{\infty} \frac{3^{-\lfloor x \rfloor} \cos(\lfloor x \rfloor \pi)}{2\lfloor x \rfloor + 1} dx = \frac{\sqrt{3}}{6}\pi$$

3. $$\int_0^{+\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$$

4. $$\int_0^3 2x \max\left(\lfloor (x^2-1)\lfloor x^2 - 1 \rfloor \rfloor, x\right) dx = 150 - \frac{1}{2} + 2\sqrt{3}.$$

Meia-final 2 - Ronda de desempate

1. $$\int_0^2 \left(3 + (x-1)^4 \sin\left((x-1)e^{(x-1)^2}\right)\right) dx = 6$$

2. $$\int_0^{\pi/2} \frac{\sqrt{\sin(x)}}{\sqrt{\sin(x)} + \sqrt{\cos(x)}} dx = \frac{\pi}{4}$$